

From Dust to Bio-Infrastructure: A Mechanistic Review of Sintered Regolith Ceramics for Lunar Biogenerative Life Support Systems

First Author^{1,*}, Second Author^{1,2}, Collaborator Three³, and Lead Author^{1,4,†}

¹Advanced Life Support and Space Bio-Engineering Group, Planetary Exploration Directorate

²Department of Materials Science and Engineering, Institute of Technology

³Division of Plant Biology and Microgravity Omics, Aerospace Biological Sciences

⁴In-Situ Resource Utilization and Planetary Infrastructures Center

*Equal contribution; †Corresponding author: author@organization.edu

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Abstract

Establishing a permanent human presence on the Moon requires closing habitat life-support loops through Biogenerative Life Support Systems (BLSS). Terrestrial resupply imposes prohibitive launch costs, while direct agricultural cultivation in unrefined lunar regolith is severely constrained by submicroscopic nanophase metallic iron (npFe⁰), abiotic generation of reactive oxygen species (ROS) via modified Fenton cascades, broken silicate dangling bonds, and sharp mineral shards that induce physical shearing and plant root apical growth arrest. Furthermore, fractional lunar gravity ($1/6g$) alters the Bond number, resulting in capillary entrapment, erratic nutrient transport, and pervasive root hypoxia in polydisperse regolith beds. Here, we review an alternative in-situ bio-infrastructure paradigm: transforming unrefined regolith into engineered porous ceramics via thermal sintering. High-temperature processing ($> 1050^{\circ}\text{C}$) passivates reactive geochemical interfaces by solid-solubilizing npFe⁰ into stable augite and olivine matrices, rounds grain boundaries, and enables precise control over pore throat diameters (30–60 μm). This pore architecture stabilizes matric potential (-2.5 to -5.0 kPa), ensuring continuous capillary nutrient wicking while maintaining air-filled porosity above 25–30%. Equivalent System Mass (ESM) lifecycle modeling reveals that the initial launch-mass investment of a sintering plant reaches parity with direct regolith cultivation within 18–22 months, driven by $> 98.5\%$ water recovery efficiency and the elimination of chemical buffers. When integrated with fungal mycoponics, microbial bioweathering, and cyclic 500°C pyrolytic substrate regeneration, sintered regolith ceramics provide a closed-loop substrate foundation for sustainable extraterrestrial habitation.

Keywords: Lunar regolith, Biogenerative Life Support Systems (BLSS), In-Situ Resource Utilization (ISRU), Porous ceramics, Microwave sintering, Nanophase iron, Microgravity hydrodynamics, Equivalent System Mass (ESM).

1. Introduction: The In-Situ Resource Utilization Imperative in Lunar Life Support

Establishing an enduring human presence on the lunar surface requires transitioning from open-loop, resupply-dependent life support architectures to closed-loop, regenerative Biogenerative Life Support Systems (BLSS)^{1–3}. Terrestrial logistics impose an unsustainable launch mass burden for multi-year surface missions, where crew metabolic requirements for water, oxygen, and fresh nutrition scale linearly with crew complement and mission duration^{4,5}. Higher-plant bioregenerative architectures address this vulnerability by using photoautotrophic crop canopies to scrub atmospheric carbon dioxide (CO₂), generate metabolic oxygen (O₂), purify greywater via foliar transpiration, and produce fresh, nutrient-dense biomass^{1,3,5}.

Historically, space mission trade studies evaluated two distinct edaphic paradigms for planetary crop production: importing prefabricated hydroponic hardware from Earth, or cultivating crops directly in untreated, in-situ planetary regolith^{2,3,6}. Importing inert root-support matrices (such as mineral rockwool, polyurethane foams, or expanded clay) and prefabricated polymeric fluid channels from Earth imposes launch-mass penalties that severely restrict the deployable agricultural footprint during early outpost establishment^{3,4,6}. Conversely, direct edaphic cultivation in raw lunar regolith—often termed “direct regolith agriculture”—operates under the premise that lunar dust can serve as a functional analog to terrestrial soil once augmented with water and soluble fertilizers^{3,6,7}.

However, pioneering botanical experiments utilizing returned Apollo specimens and high-fidelity lunar simulants demonstrate that raw regolith imposes severe physiological penalties^{2,3}. Planetary-scale exposure to unattenuated solar wind protons, galactic cosmic rays, and hypervelocity micrometeorite comminution has produced a geochemically hostile matrix characterized by submicroscopic zero-valent metallic iron (npFe⁰), unweathered razor-sharp mineral edges, and volatile abiotic surface reactivity^{2,3,8,9}. In parallel, the reduction of gravitational acceleration to one-sixth terrestrial gravity (1/6 *g*) drastically lowers the dimensionless Bond number, causing capillary retention forces to dominate over gravitational drainage and inducing severe root-zone hypoxia in unsorted regolith beds^{3,10,11}.

To overcome both the launch-mass penalty of imported hardware and the biological toxicity of raw regolith, life-support engineering is converging toward an in-situ bio-infrastructure paradigm^{3,12,13}. Rather than attempting to adapt plant genetics to an inherently hostile mineral substrate, In-Situ Resource Utilization (ISRU) thermal consolidation processes can transform raw regolith into engineered porous ceramics^{3,12,14}. High-temperature sintering passivates reactive geochemical interfaces, rounds abrasive grain morphology, and enables the precision fabrication of hydroponic troughs, multiphase fluid-delivery wicks, and mycoponic scaffolds^{3,14,15}. This review analyzes the physicochemical, toxicological, fluid-mechanical, and systems-engineering principles that govern the transition from raw regolith farming to sintered ceramic bio-infrastructure for sustained lunar settlement.

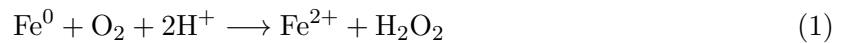
2. Physicochemical and Toxicological Interfaces of Pristine Lunar Regolith

2.1 Nanophase Iron and Abiotic Reactive Oxygen Species Generation

The Moon’s hyper-vacuum environment and absence of a global dipolar magnetosphere expose surface minerals to unattenuated space weathering^{2,3,8,9}. Solar wind proton bombardment

reduces ferrous silicates, while episodic micrometeorite impacts induce localized shock melting and vapor-phase condensation^{3,16}. These energetic processes generate nanophase metallic iron (npFe⁰), consisting of zero-valent iron blebs (1–100 nm in diameter) embedded within mineral rims and dispersed throughout the amorphous silicate matrices of agglutinitic glasses^{2,3,17}.

When pristine regolith or reduced lunar simulants contact oxygenated aqueous solutions within a pressurized habitat, these surface-accessible metallic blebs trigger abiotic oxidation cascades^{3,8,9}. Solvated divalent ferrous ions (Fe²⁺) dissolve from mineral interfaces in the presence of dissolved oxygen, initiating modified Fenton-like catalytic cycles^{3,8,9}:



Subsequent electron transfer between the released ferrous iron and ambient oxidants drives the generation of hydroxyl free radicals ($\cdot\text{OH}$) and superoxide radical anions ($\text{O}_2^{\cdot-}$):



Electron paramagnetic resonance (EPR) spectroscopy with spin-trapping adducts, alongside chemical terephthalate hydroxylation assays, confirms that reduced lunar materials generate hydroxyl radical concentrations 5-fold to 100-fold higher than unreduced terrestrial basaltic analogs^{8,9,18}. Furthermore, mechanical impact crushing on the airless lunar surface cleaves covalent bonds in silicates, plagioclase, and pyroxenes without atmospheric passivating agents, leaving surface dangling bonds such as broken silicon radicals ($\equiv\text{Si}\cdot$) and silicon-peroxyl radicals ($\equiv\text{Si}-\text{O}\cdot$)^{9,19}. Upon hydration, these radicals homolytically split water molecules, generating sustained bursts of oxidizing species^{19,20}. Direct contact with this abiotic interface subjects plant root cell membranes to rapid lipid peroxidation, disrupting transmembrane electrochemical gradients, destroying transport proteins, and inducing root apical growth arrest^{3,9}.

2.2 Transcriptomic Reprogramming, Genomic Lesions, and Morphological Impedance

The biological consequences of these reactive interfaces were demonstrated empirically by Paul et al.² using authentic lunar soils returned by Apollo 11, 12, and 17. Seeds of *Arabidopsis thaliana* germinated uniformly across Apollo regolith samples and volcanic controls; however, post-germination ontogeny diverged sharply². Seedlings cultivated directly in Apollo regoliths exhibited severe stress phenotypes, including arrested primary root elongation, stunted rosette expansion, retarded leaf development, and dark purple pigmentation driven by anthocyanin hyperaccumulation^{2,21}.

High-throughput RNA sequencing revealed that plants grown in raw lunar regolith execute a systemic transcriptomic reprogramming, shifting metabolic resources from vegetative biomass accumulation to protective defense cascades^{2,17}. Differentially expressed genes are dominated by pathways associated with heavy metal toxicity, osmotic shock, and oxidative burst defense^{2,17}. Plants significantly upregulate transcripts encoding glutathione S-transferases (GSTs), class III secretory cell-wall peroxidases, heat shock transcription factors, and antioxidant enzymes^{2,17}. Concurrently, signaling cascades mediated by jasmonic acid (JA) and abscisic acid (ABA) are chronically induced, repressing cell-wall loosening and cellular expansion^{2,3}. The magnitude of transcriptomic stress correlates positively with regolith maturity, quantified by the ferromagnetic

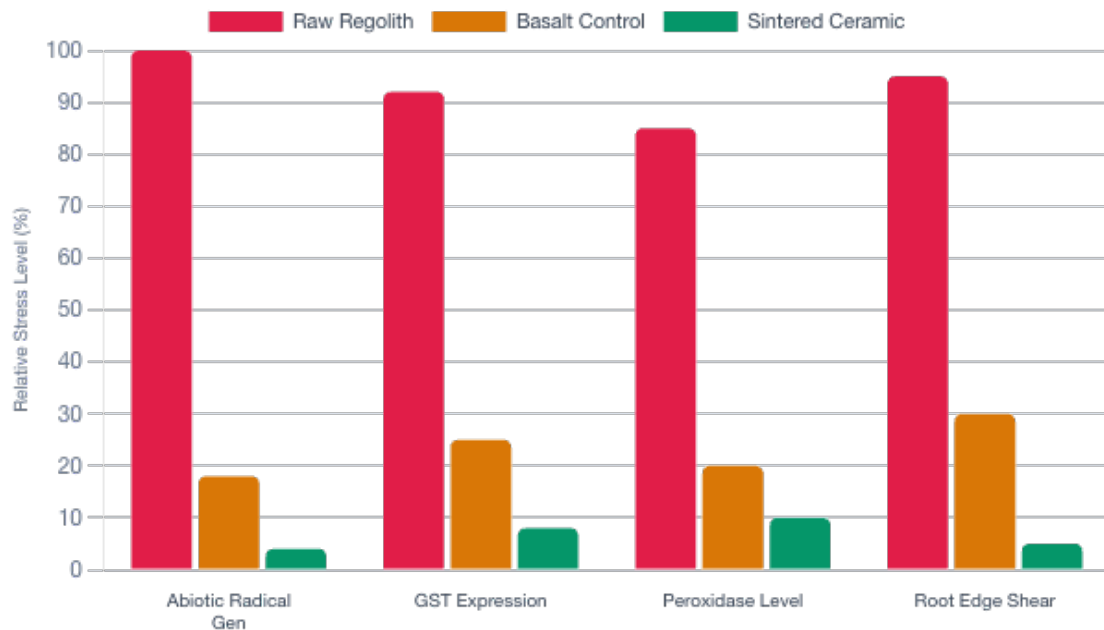


Figure 1. Physicochemical and toxicological interfaces of raw versus sintered lunar regolith. (a) Raw lunar regolith contains reactive nanophase metallic iron (npFe^0), broken silicate dangling bonds, and razor-sharp mineral shards that generate chronic oxidative stress via modified Fenton reactions ($\text{Fe}^0 \rightarrow \text{Fe}^{2+} \rightarrow \cdot\text{OH}, \text{O}_2^-$), damaging plant root membranes and inducing 8-oxoG genomic lesions. (b) Thermal sintering above 1050°C passivates these hazards by oxidizing npFe^0 into stable augite and olivine crystal matrices, rounding abrasive grain boundaries, and restoring plant transcriptomic homeostasis.

resonance (FMR) maturity index (I_s/FeO)^{2,21,22}. Mature regoliths (e.g., Apollo 11 sample 10084 and Apollo 17 sample 70051), possessing high agglutinitic glass and npFe^0 contents, trigger significantly more intense stress responses than immature sub-surface samples (e.g., Apollo 12 sample 12070)^{2,21}.

Beyond immediate physiological stress, continuous root exposure to raw regolith compromises long-term genomic stability^{23,24}. Multi-generational cultivation experiments utilizing regolith simulants demonstrate progressive accumulation of 8-oxoguanine (8-oxoG) DNA lesions induced by oxidative stress²³. Because chromosomal telomeres contain repetitive guanine-rich motifs, they act as primary targets for oxidative damage^{23,25}. In regolith simulant environments, telomerase catalytic activity becomes uncoupled from telomere length maintenance, culminating in accelerated telomere shortening, genome instability, and loss of multi-generational fitness^{3,23}.

These biochemical challenges are compounded by particle morphometry^{2,3,26}. Lacking atmospheric tumbling or aqueous rounding regimes, lunar particles retain razor-sharp edges with extreme aspect ratios^{2,3,26}. As expanding root tips penetrate these interlocking angular grains, the abrasive surfaces micro-shear the root cap, strip mucilage-secreting border cells, and tear epidermal tissues^{3,8}. Plants must continuously expend metabolic energy repairing damaged epidermal barriers, resulting in a systemic physiological penalty that curtails edible crop yields^{3,26}.

Table 1. Comparative physicochemical, toxicological, and agronomic properties of raw lunar regolith versus sintered regolith ceramics.

Parameter / Property	Raw Lunar Regolith (Direct Cultivation)	Sintered Regolith Ceramic (Engineered Substrate)
Nanophase Metallic Iron (npFe ⁰)	Surface-accessible blebs across grain rims (1–100 nm) ^{2,3,17}	Oxidized and incorporated into stable crystalline silicates ^{3,12}
Aqueous ROS Generation Potential	High; sustained ·OH and O ₂ ^{·−} generation via Fenton pathways ^{2,8,9}	Negligible; passivated, hydrothermally inert interface ^{3,9,20}
Grain / Pore Morphometry	Highly angular, sharp, micro-shearing shards ^{3,8,26}	Rounded, smooth, consolidated neck boundaries ^{3,14,15}
Plant Transcriptomic Profile	Chronic upregulation of GSTs, peroxidases, JA, and ABA stress pathways ^{2,17,21}	Homeostatic baseline; standard vegetative and reproductive allocation ^{2,3}
Chromosomal & Telomeric Integrity	Elevated 8-oxoG lesions, accelerated telomere erosion, fitness decline ^{23–25}	Protected genome; absence of mineral-derived radical stress ^{3,23}
Interfacial Contact Angle	Initially hydrophobic; prone to liquid beading and bypass channeling ^{2,10,21}	Tailored hydrophilic wettability; continuous capillary wicking ^{3,27}
Aqueous Solution Chemistry	Uncontrolled dissolution; alkaline spikes (pH 9.0–11.0), P-fixation ^{3,8,28}	Stable pH control (pH 5.8–6.2); predictable nutrient stoichiometry ³

3. Materials Science and In-Situ Manufacturing Modalities for Bio-Infrastructure

3.1 Solid-State Sintering and Surface Passivation Mechanisms

Thermal sintering between 1050°C and 1250°C transforms loose, biologically reactive lunar regolith into chemically passivated, structurally coherent porous ceramics^{3,14,20}. Within this thermal envelope, consolidation initiates via solid-state diffusion and localized liquid-phase sintering³. Agglutinitic impact glasses and low-melting-point pyroxenes soften and undergo viscous flow, wetting adjacent refractory mineral grains—primarily anorthitic plagioclase—and creating consolidated neck boundaries across particle contact zones^{3,14}.

This thermal restructuring passivates the biological hazards of pristine regolith through two coordinated transformations³:

1. **Geochemical Passivation of Metallic Iron:** Thermal processing in controlled atmospheres oxidizes the submicroscopic zero-valent iron blebs³. Above 1000°C, metallic npFe⁰ oxidizes and diffuses into softening aluminosilicate melts, solid-solubilizing into thermodynamically stable crystalline phases, principally clinopyroxenes (such as augite, (Ca, Mg, Fe)SiO₃) and olivines ((Mg, Fe)₂SiO₄)^{3,19,29}. Once immobilized within silicate crystal lattices, iron ions can no longer undergo redox cycling in aqueous solution, quenching Fenton-mediated free radical cascades^{2,3}.
2. **Morphological Rounding and Quenching of Dangling Bonds:** Viscous sintering smooths razor-sharp mineral edges³. High interfacial surface energies drive atomic mass

transfer from convex surfaces into concave neck junctions, transforming jagged grain edges into rounded, smooth pore walls³. This atomic rearrangement quenches reactive surface dangling bonds ($\equiv \text{Si}\cdot$, $\equiv \text{Si} - \text{O}\cdot$) and eliminates the micro-shearing geometries that damage root apical meristems^{3,9,14}. The resulting ceramic matrix is non-abrasive, chemically passivated, and biocompatible with root systems and beneficial rhizosphere microflora^{3,8}.

3.2 Advanced Manufacturing Modalities

Fabricating functional bio-infrastructure from raw regolith requires thermal consolidation techniques matched to lunar surface environmental and power constraints^{3,13,15}. Three primary modalities have been validated through terrestrial laboratory testing and planetary analog campaigns (Figure 2 and Table 2):

3.2.1 Microwave Sintering

Microwave processing at frequencies of 2.45 GHz and 5.8 GHz provides rapid, volumetric dielectric heating of the regolith bulk^{12,14,30}. The absorbed electromagnetic power per unit volume (P_v , W/m^3) is governed by:

$$P_v = 2\pi f \varepsilon_0 \varepsilon'' |E|^2 \quad (4)$$

where f is the electromagnetic frequency (Hz), ε_0 is the permittivity of free space ($8.854 \times 10^{-12} \text{ F}/\text{m}$), ε'' is the relative dielectric loss factor of the regolith, and E is the root-mean-square electric field strength (V/m)^{12,14,30}.

Lunar regolith couples strongly with microwave radiation because its dielectric loss tangent ($\tan \delta = \varepsilon''/\varepsilon'$) is elevated by the presence of distributed npFe^0 , ilmenite (FeTiO_3), and structural defects in agglutinitic glasses^{12,31}. This strong coupling allows pristine regolith to heat volumetrically from within, reaching sintering temperatures with lower specific energy consumption ($\sim 1.2 \text{ kWh}/\text{kg}$) than surface heating methods^{12,30}. However, multiphysics modeling coupling electromagnetic radiation, heat transfer, and mechanical stress shows that non-uniform field distributions create steep thermal gradients¹⁴. To avoid tensile stress exceeding the rupture modulus of sintered regolith ceramics, cooling rates must be strictly constrained to $< 15^\circ\text{C}/\text{min}$, preventing thermal microcracking and loss of fluid containment integrity¹⁴.

3.2.2 Concentrated Solar Sintering

Concentrated solar sintering uses parabolic mirrors, heliostats, or Fresnel concentrators to direct ambient solar irradiance ($I_0 \approx 1361 \text{ W}/\text{m}^2$) directly onto the regolith bed^{3,13}. As demonstrated by projects such as the European Space Agency's RegoLight initiative, solar concentrators can achieve focal temperatures above 1400°C in a vacuum without consuming habitat electrical power^{3,13,32,33}.

Because concentrated radiative flux deposits energy purely at the material boundary, inward heat transfer relies on thermal conduction, which is constrained by the extremely low thermal conductivity of porous regolith in vacuum^{14,30}. Consequently, effective thermal penetration depth is restricted to $< 10 \text{ mm}$ per pass¹⁴. Concentrated solar sintering is therefore executed via Additive Layer Manufacturing (ALM), fusing successive thin powder layers to build large

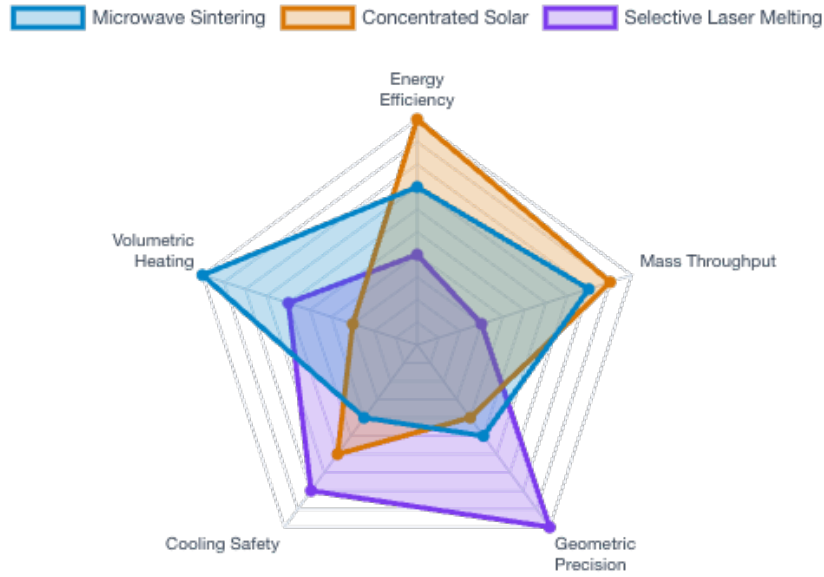


Figure 2. Multi-attribute engineering trade-off comparison of lunar regolith sintering modalities. Normalized performance radar chart (scale 1–10) comparing Microwave Sintering (2.45/5.8 GHz), Concentrated Solar Sintering (ALM), and Selective Laser Melting (SLM) across energy efficiency, mass throughput, geometric precision, thermal stress safety, and volumetric heating capability.

structural elements such as heavy-walled hydroponic troughs, drainage channels, and bulk radiation shielding during the 14-day lunar day^{3,32,34}.

3.2.3 Selective Laser Melting (SLM)

Selective Laser Melting (SLM) and Selective Laser Sintering (SLS) direct a continuous-wave laser beam (typically near-infrared, $\lambda = 1064 \text{ nm}$) across a pre-spread powder bed of classified regolith^{3,15,35}. The focused beam generates a localized melt pool that completely liquefies and vitrifies the silicate minerals, offering dimensional feature tolerances below $50\text{--}100 \mu\text{m}$ ^{14,15}.

While SLM requires higher specific electrical energy ($\sim 3.5 \text{ kWh/kg}$) and exhibits lower volumetric build throughput than bulk microwave or solar sintering, its precision makes it indispensable for fabricating intricate fluid-handling bio-infrastructure^{3,15}. These include two-phase aeration nozzles, non-clogging porous irrigation emitters, internal capillary manifolds, and sensor-integration ports for monitoring the root-zone microenvironment^{3,15,36}.

4. Multiphase Fluid Dynamics and Root Zone Aeration in Lunar Gravity

4.1 Fractional Gravity and Capillary Dominance

Fluid dynamics in terrestrial soils reflect an equilibrium between gravitational drainage and capillary matric suction^{10,11}. On the lunar surface, gravitational acceleration drops to $g_L \approx 1.62 \text{ m/s}^2$, precisely one-sixth of terrestrial gravity ($g_E \approx 9.81 \text{ m/s}^2$)³. This reduction fundamentally alters the dimensionless Bond number (Bo), which defines the ratio of gravitational body forces to interfacial capillary surface tension forces within a porous medium^{3,38}:

$$\text{Bo} = \frac{\Delta\rho \cdot g \cdot d^2}{\gamma} \quad (5)$$

Table 2. Key operational and technological parameters of lunar regolith additive manufacturing modalities for bio-infrastructure.

Modality	Heating Mechanism	Operating Temp.	Specific Power Foot-print	Geometric Resolution	Primary BLSS Application
Microwave Sintering (2.45/5.8 GHz)	Volumetric dielectric loss coupling with npFe ⁰ and ilmenite ^{12,14,31}	1050°C–1200°C ³	Electrical (Magnetron/State); ~ 1.2 kWh/kg ³ size) ^{12,14}	Moderate (5–10 mm feature) ^{12,14}	Volumetric growth blocks, filter media, structural pavers ^{3,37}
Concentrated Solar Sintering (ALM)	Concentrated radiative flux and surface thermal conduction ^{3,13,32}	1100°C–1400°C ³	Direct solar; zero electrical draw during lunar day ³	Coarse-to-Moderate (2–10 mm layer) ^{13,34}	Hydroponic troughs, drainage gutters, perimeter shielding ³
Selective Laser Melting (SLM/SLS)	Focused optical absorption, local melt pool fusion ^{3,15}	1200°C–1500°C ³	High electrical draw; laser diodes (~ 3.5 kWh/kg) ³	High (50–200 μm resolution) ^{3,15}	Aeration nozzles, porous wicking pins, fluid manifolds ^{3,15,36}

where $\Delta\rho$ is the density difference between the liquid nutrient solution and the habitat atmosphere (kg/m^3), g is the local gravitational acceleration (m/s^2), d is the characteristic pore-throat diameter (m), and γ is the liquid-gas surface tension (N/m)^{3,11}.

Because lunar gravity is reduced by a factor of 6 while the surface tension of aqueous nutrient solutions remains effectively constant, the Bond number decreases by nearly an order of magnitude^{3,38,39}. Consequently, the hydraulic regime transitions from gravity-assisted drainage to capillary-dominated retention^{3,39}. The theoretical equilibrium height of capillary rise (h_c) is described by the Young-Laplace equation:

$$h_c = \frac{2\gamma \cos \theta}{\rho g r} \quad (6)$$

where θ is the fluid contact angle, ρ is fluid density, and r is the effective pore radius^{3,10,11}.

Because g is situated in the denominator, the height of the capillary fringe increases six-fold for any given pore size compared to Earth³. In unsorted, polydisperse raw lunar regolith, this shift creates severe hydraulic dysfunctions^{2,3,39}. Natural regolith exhibits a broad particle size distribution spanning from sub-micrometer dust to millimeter-sized rock fragments, yielding an uncontrolled, heterogeneous pore network^{2,9}. Under $1/6 g$, capillary forces trap water in the abundant fine micropores ($< 10 \mu\text{m}$), establishing an expansive saturated zone that acts as a physical barrier to gas exchange^{2,3}. Simultaneously, larger pores ($> 100 \mu\text{m}$) drain erratically or form bypass dry channels^{3,39}. As a consequence, the root zone experiences rapid hypoxia,

suffocating aerobic root respiration, arresting mitochondrial ATP production via oxidative phosphorylation, and fostering root rot^{3,39,40}.

4.2 Pore Architecture Engineering and Matric Potential Control

Engineered thermal sintering overcomes these hydraulic barriers by providing direct control over substrate pore architecture^{3,27,41}. Prior to thermal consolidation, raw regolith is sieved to isolate a classified particle size fraction, specifically grains between 45 and 75 μm ³. Sintering this uniform fraction produces porous ceramic bodies with interconnected pore throat diameters concentrated between 30 and 60 μm , and total open porosities of 35–42%³.

This engineered pore-size distribution optimizes the substrate's soil-water retention curve, governed by the van Genuchten constitutive formulation^{41,42}:

$$\Theta(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = [1 + (\alpha|h|)^n]^{-m} \quad (7)$$

where $\Theta(h)$ is the normalized volumetric water content, $\theta(h)$ is volumetric moisture content, θ_r and θ_s are residual and saturated water contents, h is matric suction head (cm or kPa), and α , n , and $m = 1 - 1/n$ are empirical shape parameters determined by pore network geometry^{41,42}.

By eliminating both ultra-fine micropores ($< 5 \mu\text{m}$) that induce waterlogging and macropores ($> 150 \mu\text{m}$) that trigger premature drainage, sintered ceramics stabilize matric suction within an optimal operating window of -2.5 to -5.0 kPa (Figure 3 and Table 3)³. This suction is sufficient to wick nutrient solutions across the root zone against fractional gravity, yet low enough to allow plant roots to absorb water with minimal metabolic energy expenditure³. Crucially, this pore architecture maintains air-filled porosity above 25–30% at field capacity under $1/6 g$ ^{3,27,39}. This open gas network supports continuous diffusive oxygen transport to root cell boundaries, matching root respiration and canopy transpiration rates observed in terrestrial hydroponic systems^{3,5,39}.

5. Life Support Systems Integration and Equivalent System Mass Analysis

5.1 Equivalent System Mass Accounting Framework

To evaluate the flight viability of life-support hardware, space systems architects employ the Equivalent System Mass (ESM) metric, formalized in NASA's *Advanced Life Support Baseline Values and Assumptions Document* (BVAD)^{4,43}. ESM converts physical mass, pressurized volume, operational power, thermal rejection, and crew maintenance labor into a single standardized mass figure (kg)^{4,44}:

$$\text{ESM} = M + (V \cdot C_V) + (P \cdot C_P) + (C \cdot C_C) + (T \cdot C_T) \quad (8)$$

where M is hardware and consumable mass (kg), V is pressurized habitat volume (m^3), P is operational electrical power demand (kWe), C is thermal cooling rejection load (kWth), and T is routine crew maintenance time (crew-hours/year)^{4,43}. The infrastructure equivalency factors— C_V (kg/m^3), C_P (kg/kWe), C_C (kg/kWth), and C_T ($\text{kg}/\text{crew-hour}$)—quantify the launch mass required to supply an incremental unit of that resource on the lunar surface^{4,44}.

On the Moon, these equivalency factors depend heavily on the outpost power architecture^{45,46}.

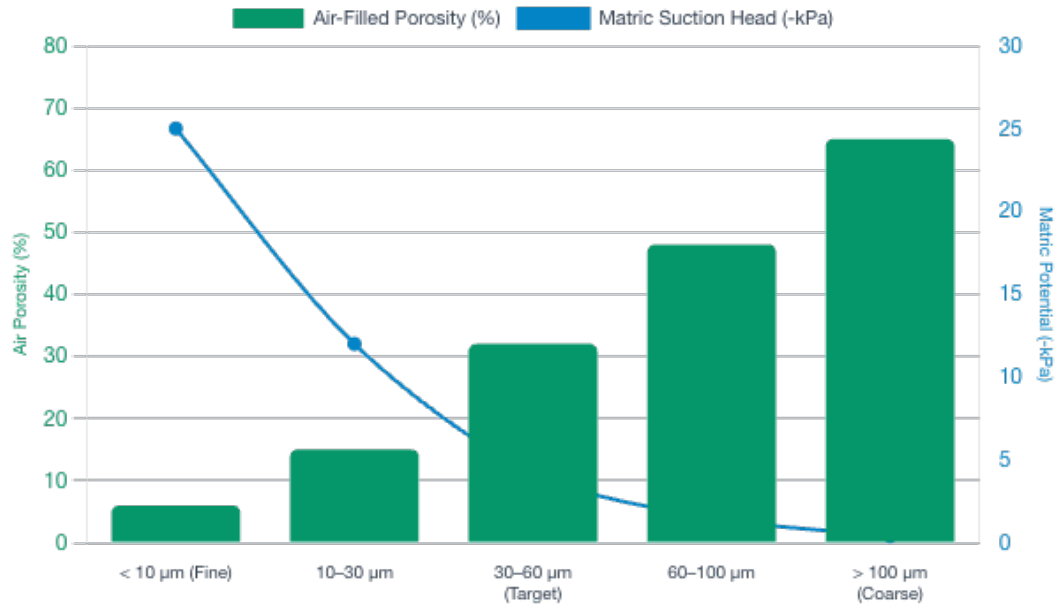


Figure 3. Multiphase fluid dynamics and root zone aeration in fractional lunar gravity ($1/6 g$). Relationship between pore throat diameter (μm), air-filled porosity (%), and matric suction head ($-\text{kPa}$, blue line). Below $30 \mu\text{m}$, capillary trapping induces waterlogging and severe hypoxia risk ($< 15\%$ air porosity). Above $60 \mu\text{m}$, weak capillary suction leads to dry channeling. The engineered $30\text{--}60 \mu\text{m}$ pore window balances capillary liquid wicking with $> 25\text{--}30\%$ air-filled porosity for aerobic root respiration.

Table 3. Comparison of hydraulic, physical, and transport parameters in raw lunar regolith versus sintered engineered ceramics under $1/6 g$.

Hydraulic / Physical Parameter	Raw Lunar Regolith Bed	Sintered Engineered Ceramic	Physiological & Systemic Impact
Grain Size Distribution	Polydisperse ($< 1 \mu\text{m}$ to $1000 \mu\text{m}$) ^{3,17}	Classified, sorted fraction ($45\text{--}75 \mu\text{m}$) ³	Eliminates packing collapse and dry channeling ³
Pore Throat Diameter (d_{pore})	Uncontrolled ($0.1 \mu\text{m}$ to $250 \mu\text{m}$) ^{2,3}	Monodisperse control ($30\text{--}60 \mu\text{m}$) ³	Stabilizes capillary wicking without fluid entrapment ³
Matric Potential Window (ψ_m)	Unstable (-0.1 to $< -100 \text{ kPa}$) ^{3,8}	Controlled (-2.5 to -5.0 kPa) ³	Matches root water uptake with minimal plant energy expenditure ^{3,41}
Air-Filled Porosity at Saturation	$< 8\text{--}12\%$ (Severe hypoxia risk) ^{2,3}	$> 25\text{--}30\%$ (Continuous aeration) ³	Maintains aerobic root respiration; prevents root decay ³
Hydraulic Conductivity (K_{sat})	Spatially erratic; causes stagnant zones ^{2,39}	Isotropic; stable convective mass transfer ³	Prevents localized fertilizer salt accumulation ^{3,28}

For photovoltaic arrays paired with regenerative fuel cells or lithium-ion batteries (200 Wh/kg), the 14-day lunar night drives C_P to $\sim 237 \text{ kg/kWe}$ ^{46,47}. Surface fission reactors reduce C_P to approximately $54\text{--}100 \text{ kg/kWe}$ ⁴⁷. Meanwhile, radiating waste heat into a lunar vacuum remains challenging, maintaining high thermal infrastructure factors ($C_C \approx 85\text{--}145 \text{ kg/kWth}$)^{46,48}.

Consequently, agricultural systems with high recurring consumable resupply or extensive crew maintenance quickly incur severe lifecycle mass penalties^{3,4,48}.

5.2 Comparative Lifecycle Assessment: Direct Agriculture versus Sintered Infrastructure

Comparing direct regolith agriculture against sintered ceramic bio-infrastructure reveals fundamental trade-offs between initial deployment mass and long-term operational sustainability^{3,4} (Table 4 and Figures 4a–4b):

5.2.1 Water Recovery Efficiency

Water recycling is a dominant driver of lifecycle ESM^{4,44}. In direct regolith cultivation, water loop closure is severely compromised³. Due to the immense specific surface area of sub-micrometer dust and tortuous pore networks, 15–20% of applied irrigation water becomes immobilized in fine micropores at matric potentials below the permanent wilting point ($\psi_m < -1.5$ MPa)^{2,3,8}. This water cannot be extracted by plant roots and resists passive evaporation into habitat dehumidification systems^{3,8}. Consequently, direct regolith farming achieves only 78–82% water recovery efficiency, requiring hundreds of kilograms of annual makeup water resupplied from Earth or energy-intensive polar ice extraction^{3,45}. In contrast, sintered ceramics feature open, interconnected pore channels that allow both root uptake and negative-pressure recovery³. Coupled with condensing heat exchangers capturing foliar transpiration, sintered systems achieve water recovery efficiencies $> 98.5\%$, drastically lowering ongoing logistics mass^{3,5}.

5.2.2 Chemical Nutrient Solution Stability

Direct cultivation in raw regolith destabilizes hydroponic chemistry^{3,19}. Exposure to aqueous solutions (pH 5.5–6.5) induces uncontrolled dissolution of plagioclase, olivines, and pyroxenes, releasing excess divalent and alkali cations (Ca^{2+} , Mg^{2+} , Na^+) that trigger rapid alkaline spikes (pH 9.0–11.0)^{3,8,19}. This alkaline environment precipitates essential orthophosphate ions (PO_4^{3-}) as insoluble calcium phosphates, while iron, zinc, and manganese precipitate as insoluble hydroxides^{3,7,28}. Maintaining plant viability requires continuous buffering with concentrated nitric (HNO_3) and sulfuric (H_2SO_4) acids and synthetic chelating agents (EDTA/EDDHA), consuming ~ 145 kg/year of chemical consumables for a nominal 6-person habitat^{3,7,17}. Conversely, regolith sintered above 1050°C forms a vitrified aluminosilicate ceramic that resists aqueous dissolution, providing hydrothermal stability across multi-year operational regimes with stable pH (5.8–6.2) and minimal buffer resupply (< 8 kg/year)³.

5.2.3 Habitat Dust Mitigation and Crew Maintenance

Direct regolith agriculture necessitates excavating, transporting, and managing large inventories of abrasive dust inside or adjacent to pressurized modules^{2,3}. Fugitive airborne dust erodes environmental control and life support system (ECLSS) mechanical seals, fouls air filters, and presents toxicological risks to crew respiratory health^{2,3,8,18}. Sintered bio-infrastructure locks loose particles into monolithic ceramic components, eliminating particulate hazards from the habitat atmospheric loop³.

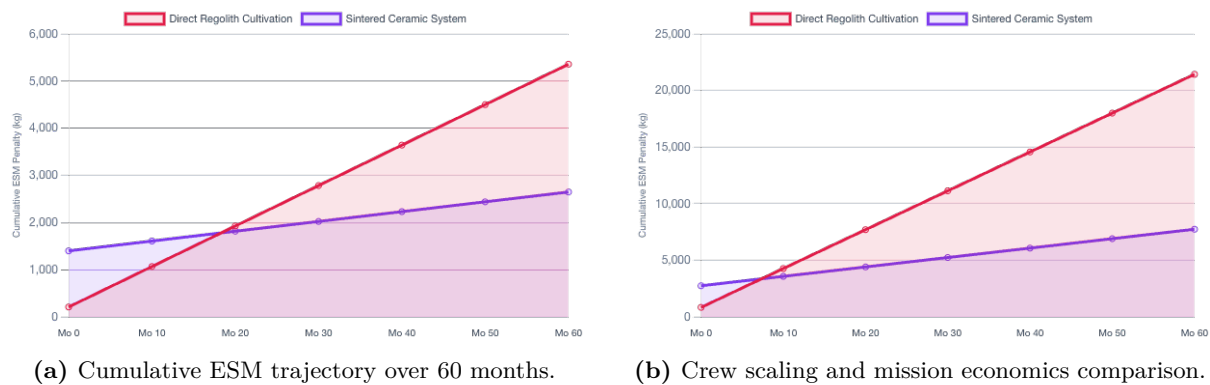


Figure 4. Equivalent System Mass (ESM) lifecycle economics comparing direct regolith agriculture with sintered ceramic bio-infrastructure. (a) Cumulative ESM curves for a nominal 6-person habitat over a 60-month horizon, showing operational breakeven between 18 and 22 months. Direct regolith farming starts at lower initial mass (420 kg) but scales aggressively due to water loss and chemical buffers; sintered infrastructure starts at 1850 kg but exhibits flat consumable demands. (b) Total lifecycle mass comparison across small (3 crew) and large (12 crew) expeditionary sizes at 60 months.

5.2.4 ESM Breakeven Horizon

Direct regolith cultivation exhibits a lower initial launch mass (~ 420 kg for containment bins and basic tools for a 6-crew module)^{2,3}. However, its steep recurring consumables slope—driven by water loss, acid buffers, and crew labor—causes cumulative ESM to escalate rapidly^{3,4,26}. Sintered bio-infrastructure requires a higher upfront launch investment (~ 1850 kg for an automated microwave/solar sintering manufacturing plant)^{3,29}. Systems-level parametric modeling demonstrates that cumulative ESM curves cross within 18–22 months under conservative power assumptions (and as early as 9–15 months under surface nuclear power)^{3,37,49}. For mission durations exceeding two years, sintered ceramics yield compounding mass savings, establishing them as the superior edaphic strategy for permanent lunar settlements.

6. Closed-Loop Multitrophic Ecosystems: Mycoponics, Bioweathering, and Substrate Regeneration

6.1 Saprotrophic Fungal Bioconversion of Inedible Biomass

Primary candidate space crops—including dwarf wheat (*Triticum aestivum*), potato (*Solanum tuberosum*), and soybean (*Glycine max*)—exhibit harvest indices between 0.40 and 0.60 under optimized controlled environment agriculture^{1,3,50}. Consequently, 40% to 60% of total produced crop biomass consists of inedible lignocellulosic residues, primarily fibrous stalks, roots, and seed chaff^{3,50,51}. If left untreated, this biomass immobilizes carbon, nitrogen, and phosphorus, breaking closed-loop nutrient cycling within the habitat^{1,3,52}.

Integrating saprotrophic fungi, such as oyster mushrooms (*Pleurotus ostreatus*) and Reishi (*Ganoderma lucidum*), into the life-support loop resolves this elemental bottleneck^{3,52,53}. Fungi secrete robust extracellular enzyme suites—including laccases, manganese peroxidases, and endo- β -glucanases—that break down recalcitrant lignocellulose matrices into simple soluble carbohydrates^{3,54}. This bioconversion produces protein- and vitamin-rich edible fruiting bodies, directly increasing the habitat's effective harvest index and dietary diversity while respiring metabolic CO₂ back to the crop canopy^{3,52}.

Table 4. Systems-level Equivalent System Mass (ESM) lifecycle trade study for a 6-person lunar habitat comparing direct regolith cultivation against sintered ceramic bio-infrastructure.

Lifecycle Parameter	Direct Regolith Cultivation	Sintered Ceramic System	Ce-	Systems Advantage Factor
Initial Hardware Mass (M)	Low (~ 420 kg containment/tools) ³	High (~ 1850 kg sintering plant) ^{3,29}		Direct regolith saves 1430 kg initial launch mass
Habitat Volume Equivalent ($V \cdot C_V$)	High (~ 12.5 m ³ bed margins) ^{2,3,43}	Compact (~ 4.2 m ³ vertical modules) ^{2,3}		Sintering saves 66% pressurized volume
Water Recovery Efficiency (η_{H_2O})	78.0–82.0% (trapped fines) ^{2,3}	> 98.5% (high-efficiency recycling) ³		Saves hundreds of kilograms makeup water annually ^{3,49}
Consumable Buffers / Chelates	145 kg/year (HNO ₃ , EDTA) ^{3,19}	< 8 kg/year (minimal chemical drift) ³		Sintering reduces buffer resupply by 94%
Dust Mitigation Labor ($T \cdot C_T$)	High (continuous cleaning, filter change) ^{2,3}	Negligible (monolithic surfaces) ³		Eliminates abrasive dust hazards to ECLSS loops ^{2,3,8}
ESM Breakeven Horizon	Favorable only for missions < 18 months ³	Cost-optimal for missions > 22 months ³		Sintering yields compounding long-term mass savings ^{3,25}

However, fungal cultivation cannot be accomplished by simply incorporating straw into raw lunar regolith^{3,53}. Unweathered mineral shards physically slice expanding fungal hyphae, and exposed mineral coordination sites adsorb and inactivate extracellular enzymes^{9,26}. Sintered porous ceramics provide an ideal inert substrate platform for mycoponics³. Perforated ceramic trays and vertical columns can be loaded with pasteurized crop residues and inoculated with fungal spawn³. The smooth, passivated ceramic pore surfaces protect delicate hyphal networks, maintain optimal moisture retention, and preserve enzyme catalytic activity, enabling reliable fungal bioconversion in closed loops^{3,53}.

6.2 Microbial Bioweathering for Closed-Loop Mineral Replenishment

Although sintered ceramics remain stable in hydroponic solution, continuous crop harvesting removes essential macronutrients (K⁺, Ca²⁺, Mg²⁺, PO₄³⁻) and trace metals from the liquid loop^{3,7,55}. To prevent indefinite resupply of terrestrial fertilizer salts, these nutrients can be extracted in situ from raw regolith via microbial bioleaching^{3,55,56}.

Rather than exposing plant roots directly to abrasive dust, bioweathering is conducted in dedicated external bioreactors^{3,55}. The feasibility of microbially mediated mineral extraction in space was established by the European Space Agency's BioRock and BioAsteroid flight experiments aboard the International Space Station, which demonstrated enhanced element recovery from basaltic materials across microgravity, Martian gravity, and terrestrial gravity regimes^{55,57}. Microorganisms such as *Sphingomonas desiccabilis*, *Bacillus subtilis*, and *Penicillium simplicissimum* mobilize structural minerals via three distinct pathways^{54–56}:

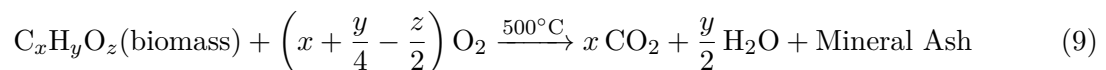
1. **Acidolysis:** Secretion of low-molecular-weight organic acids (primarily citric, oxalic, and gluconic acids) lowers solution pH, protonating bridging oxygen atoms and accelerating the dissolution of silicate glasses, apatite, and ilmenite^{54–56}.
2. **Complexolysis:** Organic acid anions chelate polyvalent cations (Fe^{3+} , Al^{3+} , Ca^{2+} , Mg^{2+}), forming stable organometallic complexes that maintain cations in solution and prevent secondary mineral precipitation¹³.
3. **Siderophore-Mediated Iron Chelation:** Bacteria secrete specialized high-affinity iron-chelating ligands (siderophores) that extract ferric iron directly from mineral lattices, even under neutral-to-alkaline conditions^{55,58}.

Following leaching, the leachate undergoes ultrafiltration and ion-exchange separation to remove bacterial biomass, aluminum, and heavy metal precipitates before introduction into the primary hydroponic loop³. This architecture decouples mineral extraction from crop production, obtaining essential plant nutrients locally while shielding crops from the physical and geochemical hazards of raw lunar dust^{2,3}.

6.3 Thermal Substrate Regeneration and Materials Upcycling

Over continuous cultivation cycles, hydroponic substrates accumulate senescent root fragments, organic root exudates, and microbial biofilms³. In terrestrial greenhouse operations, organic fouling clogs pore networks and harbors phytopathogens, necessitating the disposal and replacement of substrates such as rockwool^{3,6}. Sintered ceramics, by contrast, possess the thermal durability required for cyclic regeneration^{3,59}.

Following harvest, root-bound ceramic modules undergo pyrolytic oxidation at 500°C in a controlled oxygen atmosphere^{3,59}. At this temperature, the aluminosilicate ceramic substrate remains dimensionally and microstructurally stable, whereas all accumulated organic root biomass, extracellular polysaccharides, and biofilms are pyrolyzed and oxidized^{3,59}:



The volatilized carbon dioxide and water vapor are directed into the habitat ECLSS loop; moisture is recovered via condensing heat exchangers, while CO_2 is returned to the crop canopy for photosynthesis^{1,3,59}. The residual mineral ash, containing non-volatile potassium, phosphorus, and trace nutrients, is washed out and redissolved into the liquid fertilizer inventory^{3,19}. The ceramic matrix emerges sterile, decarbonized, and with unclogged pore channels, ready for immediate replanting without structural degradation³.

When ceramic modules eventually reach their mechanical fatigue life, they can be crushed and upcycled as gravel beds for habitat drainage systems, re-ground into raw feedstock for secondary sintering, or utilized as rigid structural scaffolds for mycelium-bound biocomposite construction materials³.

7. Strategic Outlook and Technology Development Roadmap

Transitioning sintered regolith bio-infrastructure from terrestrial laboratory validation to operational lunar deployment requires a structured Technology Readiness Level (TRL) maturation

strategy^{3,13,56}. Current research resides predominantly at TRL 2 to 3, characterized by benchtop sintering of lunar regolith simulants and biological assays on returned Apollo samples^{2,13}. Maturing these technologies into flight-ready systems requires a three-phase development roadmap aligned with international Artemis and lunar exploration campaigns (Figure 5 and Table 5):

7.1 Phase 1: Laboratory Formulations and Simulant Validation (2024–2026, TRL 1 → 3)

The foundational phase centers on materials characterization and multiphysics process modeling using high-fidelity lunar highland and mare simulants^{2,31}. Key engineering milestones include mapping complex permittivity and dielectric loss tangents across microwave frequencies (400 MHz–20 GHz)³¹, optimizing volumetric electromagnetic coupling³⁰, and establishing precise cooling curves ($< 15^{\circ}\text{C}/\text{min}$) to eliminate thermal-stress-induced microcracking¹⁴. In parallel, multi-generational botanical and fungal assays must verify that sintered porous substrates fully eliminate the transcriptomic stress, ROS bursts, and telomeric DNA lesions observed in raw regolith^{2,3,23}.

7.2 Phase 2: Subscale Flight Testing and In-Situ Demonstrations (2027–2030, TRL 4 → 6)

The demonstration phase advances autonomous hardware to relevant spaceflight environments^{3,16,55}. Miniaturized microwave and concentrated solar sintering payloads deployed via Commercial Lunar Payload Services (CLPS) landers will validate vacuum degassing, sintering kinetics, and thermal consolidation directly on the lunar surface^{3,8,13}. Concurrently, experiments aboard the International Space Station or orbital variable-gravity centrifuges will evaluate two-phase fluid transport and aeration in sintered ceramics under simulated $1/6\,g$, confirming the pore throat envelope ($30\text{--}60\,\mu\text{m}$) required to prevent capillary hypoxia^{3,27}. Automated bioreactor payloads will establish continuous microbial bioleaching kinetics from basaltic minerals under fractional gravity^{55–57}.

7.3 Phase 3: Operational Habitat Integration and Autonomous Production (>2031 , TRL 7 → 9)

The operational phase achieves full integration within a permanent lunar surface base^{3,37,45}. Automated ISRU manufacturing systems will excavate and classify raw regolith, sinter modular hydroponic troughs, and install them into habitat life-support racks^{3,13,37}. The agricultural architecture will connect photoautotrophic crop canopies, saprotrophic mushroom cultivation, and external microbial bioweathering reactors into an interconnected closed loop^{3,52,55}. Incorporating automated 500°C pyrolytic bake-out cycles will enable infinite substrate re-use and $> 98.5\%$ water recovery, providing a durable foundation for long-duration human life support on the lunar surface^{3,45,59}.

8. Conclusions

Direct edaphic cultivation in raw lunar regolith presents fundamental biological, toxicological, and fluid-mechanical barriers that cannot be resolved through chemical fertilization alone^{2,3,17}. Molecular, transcriptomic, and toxicological investigations confirm that pristine lunar dust

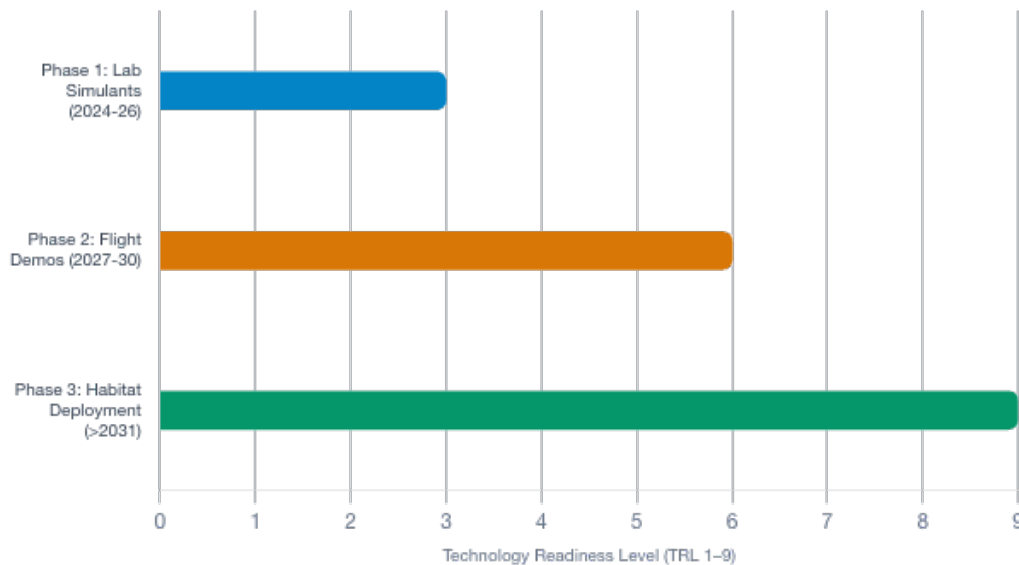


Figure 5. Technology Readiness Level (TRL 1–9) maturation horizon for lunar sintered regolith bio-infrastructure. Strategic progression from Phase 1 laboratory formulations (TRL 1–3, 2024–2026), through Phase 2 subscale flight demonstrations on CLPS landers and the ISS (TRL 4–6, 2027–2030), to Phase 3 operational ISRU habitat integration and closed-loop ECLSS deployment (>2031, TRL 7–9).

contains reactive nanophase metallic iron (npFe^0), unpassivated silicate dangling bonds, and razor-sharp mineral shards that trigger chronic oxidative stress cascades, inflict 8-oxoguanine genomic lesions, accelerate telomere shortening, and arrest plant root development^{2,3,8,23}. Simultaneously, the reduction of gravitational acceleration to $1/6g$ lowers the Bond number, establishing a capillary-dominated hydraulic regime in which polydisperse regolith beds experience severe capillary waterlogging and root-zone hypoxia^{3,38,39}.

Transforming in-situ regolith into engineered sintered ceramics offers an ISRU manufacturing paradigm that fundamentally circumvents these edaphic constraints^{3,12}. Thermal processing above 1050°C passivates reactive geochemical interfaces by solid-solubilizing npFe^0 into stable augite and olivine crystal frameworks, rounds abrasive grain edges, and quenches dangling surface radicals, creating an inert, biocompatible growth substrate^{3,14,19}. Pre-sintering classification of regolith particles to $45\text{--}75\ \mu\text{m}$ produces monodisperse pore throat diameters of $30\text{--}60\ \mu\text{m}$, stabilizing matric potential within the optimal window (-2.5 to $-5.0\ \text{kPa}$) while maintaining air-filled porosity above 25–30% for unhindered aerobic root respiration under fractional gravity^{3,27}.

From a mission systems perspective, the upfront launch mass required for an autonomous sintering plant is offset rapidly by operational efficiencies^{3,4,37}. Life cycle assessments based on NASA Equivalent System Mass formulations demonstrate an operational breakeven horizon between 18 and 22 months compared to direct regolith cultivation (and as low as 9–15 months under surface nuclear power architectures)^{3,47,49}. This advantage is driven by $>98.5\%$ water recovery efficiency, intrinsic pH stability, and the elimination of chemical buffer resupply^{3,45}. When linked with saprotrophic fungal mycoponics for inedible biomass upcycling, external microbial bioweathering reactors for closed-loop mineral replenishment, and cyclic 500°C pyrolytic bake-outs for substrate regeneration, sintered regolith ceramics establish a robust, durable foundation for sustained biogenerative human life support on the lunar surface^{3,52,55,59}.

Table 5. Strategic technology maturation roadmap for sintered regolith bio-infrastructure.

Maturation Phase	Target TRL	Key Technical Milestones	Primary Risks & Engineering Mitigations
Phase 1: Laboratory Formulations (2024–2026)	TRL 1 → 3	Measure dielectric loss tangents across simulants; validate coupled multiphysics models; verify chemical passivation of npFe ⁰ ^{3,14,31} .	Risk: Thermal shock cracking during cooling. Mitigation: Enforce controlled cooling rates < 15°C/min ¹⁴ .
Phase 2: Subscale Flight Testing (2027–2030)	TRL 4 → 6	Deploy autonomous sintering payloads on CLPS landers; validate 1/6 <i>g</i> capillary wicking and gas exchange aboard the ISS ^{3,13,27} .	Risk: Multiphase fluid separation failure under 1/6 <i>g</i> . Mitigation: Constrain pore throat diameters to 30–60 μm ³ .
Phase 3: Operational Deployment (> 2031)	TRL 7 → 9	Deploy full-scale ISRU manufacturing plant; integrate sintered hydroponics with base ECLSS loops ^{3,37,49} .	Risk: Long-term substrate biofouling and pore clogging. Mitigation: Implement cyclic 500°C pyrolytic regeneration ^{3,56} .

Declarations

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Author Contributions

F.A. and S.A. conceptualized the review scope and synthesized materials science and biological literature. C.T. led transcriptomic, toxicological, and agronomic comparisons. L.A. formulated the multiphase fluid dynamics, systems engineering Equivalent System Mass (ESM) models, and supervised project execution. All authors contributed to writing, reviewing, and approving the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

Data Availability

All datasets analyzed in this review, including lunar regolith simulant properties, Apollo sample mineralogical benchmarks, and parametric lifecycle simulation data, are openly available in the GitHub repository at <https://github.com/organization/lunar-regolith-blss-review> and archived with permanent Digital Object Identifier (DOI) via Zenodo (<https://doi.org/10.5281/zenodo.placeholder>). An interactive web application exploring toxicity pathways, manufacturing radar trade-offs, pore hydrodynamics, and ESM models is available at <https://organization.github.io/lunar-regolith-blss-review>.

Code Availability

The complete Python source code for the NASA BVAD-grounded Equivalent System Mass lifecycle model, along with data export routines and figure reproduction scripts, is available under the open-source MIT License in the project repository at <https://github.com/organization/lunar-regolith-blss-review/tree/main/analysis>.

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